Same Motion, Different Scores: Supplementary Material

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This supplement contains the equivalence proof and verification (Sec. S1), equal-budget and physical controls (Secs. S2 and S3), the full public-output audit (Sec. S4), attack protocols (Sec. S5), reward fine-tuning (Sec. S6), the sign-language experiments (Sec. S7) and statistical details (Sec. S8). Code, configurations, saved outputs and checkpoint hashes accompany the submission.

HumanML3D [1] and the KIT Motion-Language (KIT-ML) dataset [2] pair 3D human motion sequences with text descriptions. We use Guo et al.'s text-to-motion (T2M) evaluator [1] and Text-to-Motion Retrieval (TMR) [3]. R-Precision (R1 for top-1) measures retrieval of a paired caption among candidate descriptions; FID applies the Fréchet distribution distance from image evaluation [4] to motion embeddings [1].

S1. Equivalence and verification

Fiber. For $a \neq 0$ and b not parallel to a, let $u = a/\|a\|$, $b_\perp = b - (u^\top b)u$ and $v = b_\perp/\|b_\perp\|$. Under $a' = \alpha a$, $b' = \beta b + \eta a$ with $\alpha, \beta > 0$, the first vector normalizes to u and $b'_\perp = \beta b_\perp$, so v and $R = [u; v; u \times v]$ are unchanged. Conversely, every d with representative (u, v) is $a = \rho u$, $b = \sigma v + \tau u$ with $\rho, \sigma > 0$, and any two such encodings are related by one recoding. Recodings compose as $(\alpha_2\alpha_1,\ \beta_2\beta_1,\ \beta_2\eta_1 + \eta_2\alpha_1)$, with inverse $(\alpha^{-1}, \beta^{-1}, -\eta/(\alpha\beta))$. Negative scales can flip rows and are excluded.

Canonicalization. $\mathcal{C}(a, b) = (u, v)$ is idempotent and constant on fibers, so $f \circ \mathcal{C}$ is invariant for any deterministic reader f. Its derivative vanishes along $(a, 0)$, $(0, b)$ and $(0, a)$, the three fiber directions.

Numerical checks. On 4,096 random double-precision features, fiber reconstruction error is 4.6×10^{-15}, canonical invariance error 4.9×10^{-15} and directional derivatives are below 3×10^{-14}; our converter matches the official one to 4.5×10^{-16}. For every experiment, saved arrays are checked independently of any score: the maximum rotation discrepancy is below 4×10^{-5} degrees, HumanML3D position channels are bitwise unchanged, sampled forward kinematics differs by at most 9.6×10^{-7}, and all non-rotation channels and padding are bitwise identical. Because tiny floating-point differences can flip a near-tied decoder, hypothesis equality is verified on saved strings rather than inferred.

Table S1. **Equal-budget controls, $\kappa = 3$, mean of 20 draws.** Left: T2M ΔFID; right: TMR ΔR1 in percentage points. Every fixed-fiber T2M interval excludes zero; all intervals and $\kappa = 1$ are in the released tables.

	T2M ΔFID			TMR ΔR1 (pp)		
Block	random	fixed +	fixed −	random	fixed +	fixed −
Fiber	0.0024	0.0823	0.1169	+0.01	−0.10	−0.23
Rotation	0.0035	0.0106	0.0151	+0.02	−0.22	+0.01
Velocity	0.0159	0.0603	0.4760	−0.00	−0.22	−0.19
Contact	0	0	0	−0.24	−0.84	−2.32
Position	0.0183	0.3416	1.1890	+0.03	−0.18	−0.08

S2. Equal-budget block controls

For evaluator r with fixed per-channel standard deviations D_r, each intervention on a clip of L valid frames is rescaled to $\|D_r^{-1}(y - x)\|_2 = \kappa\sqrt{L}$, counting all 263 channels. *Fiber* interventions change only redundant coordinates; *rotation* interventions apply a local SO(3) change and re-encode it canonically; the other blocks receive additive changes to velocities, contacts or positions only. *Random* uses i.i.d. Gaussian directions; *fixed* uses one direction shared by all joints and frames, with both signs. Only fiber interventions keep every represented quantity fixed. The T2M evaluator does not read contacts.

Within the rotation block, the two tested fixed fiber directions change T2M FID by 7.76× and 7.73× as much as the fixed rotation directions of the same sign; at $\kappa = 1$ the ratios are 8.07 and 7.68. These are descriptive ratios for the tested directions, not a universal ordering: random fiber coefficients have smaller effects than random rotation changes,

Table S2. **FID of physical and encoding changes** against the same reconstructed real motion, mean of 20 draws.

Angle	1°	3°	10°	30°
Fixed axis	0.0001	0.0026	0.300	14.44
Per-frame axis	0.350	10.24	38.98	75.17

Recodings (no physical change): fixed ±0.1: 0.059 / 0.083; fixed ±0.35: 1.97 / 3.57; random 0.35: 0.024.

and fixed directions in non-rotation blocks can have larger effects (Tab. S1). What distinguishes the fiber is that it changes no represented quantity, so no response to it is warranted.

S3. Physical calibration

We rotate the 21 non-root joints of real HumanML3D motion by a local angle about a random axis, either fixed through the clip or redrawn per frame, and recompute rotations, positions, velocities and contacts with the native skeleton. All conditions are compared with the same zero-angle reconstruction, whose own FID to the original data is 0.020 (R1 0.451→0.447).

S4. Public-output audit

Protocol. We use the native HumanML3D test loader [1] with its caption, crop and mirroring draws (4,640 entries after dropping the last incomplete batch of 32) and 20 seeds; the real reference is an independent draw. The Human Motion Diffusion Model (MDM) [5], MotionDiffuse [6] and DiP [7] emit 1,024 outputs per draw and the others 4,640; KIT-ML uses 704. Native inference settings are kept: MDM 50 steps (CFG 2.5), MLD [8] DDIM-50 (CFG 7.5), MotionLCM [9] one step, T2M-GPT [10] categorical sampling, T2M with its length estimator, MoMask [11] 18 mask steps, MotionDiffuse 1,000-step DDPM, and DiP with a 20-frame prefix excluded from scoring. Raw and canonical scores share every output, caption, candidate batch and reference. Intervals are paired t_{19}, Bonferroni-adjusted over ten FID/R1 endpoints per study.

Second checkpoint and retrieval. Table S3 repeats the original five releases with the distinct official T2M weights distributed with T2M-GPT. Four FID gaps exclude zero there; MLD's is inconclusive and T2M's changes sign. Retrieval gaps are small with both checkpoints and with TMR (Tab. S4).

Ranking changes. For every pair of releases we compute the per-draw FID and R-Precision difference under raw and under canonical inputs, which is paired because all releases share the same 20 caption draws and reference draws, and form t_{19} intervals Bonferroni-adjusted over all pairs and both encodings (56 comparisons for the eight releases, 20 for the five scored by the second checkpoint). With the main checkpoint, the FID ranking of all eight releases is identical before and after canonicalization (Kendall $\tau = 1$). With the second checkpoint, the means of MLD and MotionLCM swap (raw difference $+0.001$, interval $[-0.027, 0.030]$; canonical -0.008, $[-0.039, 0.024]$); the change in their difference, 0.009 $[0.005, 0.013]$, excludes zero, but neither ordering is supported. For R-Precision, MDM and MLD swap in the means with both intervals containing zero. No pair has a supported ordering under one encoding and the opposite supported ordering under the other.

Essential-feature readers. The public MARDM [12] GRU and contrastive readers, which read only the 67 root and position features, are exactly invariant to rotation recoding. Their release rankings differ from that of canonical T2M by two adjacent swaps each (Kendall $\tau = 0.857$ for both).

Published margins. Table 1 of the main paper compares gaps with author-reported 95% margins. These margins describe different protocols and are used only for scale. We use MDM's corrected 0.489±0.047, T2M with estimated lengths, MoMask's final KIT table and MotionDiffuse's release preprint.

S5. Attack protocols

Targeted recoding. We sample 1,024 full-caption test clips (40–199 frames) once, with fixed captions and crops, and use them for real motion, MDM outputs and both evaluators. Per-frame, per-joint coefficients start at zero and are optimized for 80 Adam steps (learning rate 0.1, bound 0.35) to minimize embedding distance to the actual or a circularly shifted caption, with evaluator weights frozen. R-Precision averages five fixed 32-candidate permutations. Optimizing through a canonicalizing evaluator leaves every score at its baseline. The main paper uses the T2M checkpoint released with T2M-GPT, whose real-motion scores match the published ones; Tab. S5 repeats the attack with the checkpoint distributed with MDM. Attacks against one checkpoint largely do not transfer to the other (MDM outputs attacked on the MDM-distributed checkpoint reach only 0.512 on the other), as expected for a per-clip attack with gradient access. FID moves in different directions for the two checkpoints, so the attack moves retrieval specifically rather than every metric.

Caption-blind recoding. For 960 fresh MDM outputs whose source clips are disjoint from the targeted study, we compute embedding moments (μ_s, Σ_s) and those of

Table S3. **Raw minus canonical with two official T2M checkpoints**, 20 draws. Intervals adjust ten endpoints per checkpoint.

Checkpoint	Release	Raw FID	$\mathcal{C}$ FID	FID gap 95% CI	Raw R1	$\mathcal{C}$ R1	R1 gap 95% CI
MDM-linked	MDM	0.4550	0.4504	$[+0.0035, +0.0058]$	0.4115	0.4115	$[-0.0017, +0.0018]$
MDM-linked	MLD	0.4453	0.4491	$[-0.0052, -0.0024]$	0.4118	0.4105	$[+0.0003, +0.0023]$
MDM-linked	MotionLCM	0.5747	0.6062	$[-0.0347, -0.0282]$	0.4374	0.4350	$[+0.0005, +0.0043]$
MDM-linked	T2M-GPT	0.1828	0.1488	$[+0.0325, +0.0354]$	0.4226	0.4240	$[-0.0030, +0.0002]$
MDM-linked	T2M	1.3935	1.3471	$[+0.0441, +0.0486]$	0.3933	0.3930	$[-0.0009, +0.0017]$
T2M-GPT	MDM	0.5129	0.5104	$[+0.0012, +0.0040]$	0.4586	0.4584	$[-0.0015, +0.0019]$
T2M-GPT	MLD	0.4206	0.4216	$[-0.0023, +0.0002]$	0.4669	0.4666	$[-0.0012, +0.0017]$
T2M-GPT	MotionLCM	0.4195	0.4294	$[-0.0138, -0.0060]$	0.5042	0.5026	$[-0.0004, +0.0035]$
T2M-GPT	T2M-GPT	0.1204	0.1115	$[+0.0076, +0.0103]$	0.4909	0.4898	$[-0.0004, +0.0026]$
T2M-GPT	T2M	1.0836	1.0917	$[-0.0111, -0.0051]$	0.4547	0.4540	$[-0.0007, +0.0021]$

Table S4. **TMR retrieval, raw minus canonical**, 20 draws; intervals adjust twelve endpoints.

Release	Raw R1	R1 gap 95% CI
MDM	0.5578	$[-0.0015, +0.0029]$
MLD	0.5848	$[-0.0005, +0.0019]$
MotionLCM	0.6785	$[+0.0013, +0.0043]$
T2M-GPT	0.6703	$[-0.0006, +0.0017]$
T2M	0.5626	$[+0.0003, +0.0028]$

Table S5. **Caption-targeted recoding with both T2M checkpoints**, 1,024 test clips. R1: top-1 R-Precision for the true caption, and for another clip's caption (*other*). FID against the same real test clips. Real-motion R1 is 0.517 (T2M-GPT checkpoint) and 0.462 (MDM checkpoint); the published value is 0.511.

Checkpoint	Motion	R1 before	R1 after	R1 other	FID before→after
T2M-GPT	MDM	0.457	0.997	0.916	0.497→0.393
T2M-GPT	Real	0.517	0.998	0.952	0.000→0.385
MDM	MDM	0.397	0.993	0.788	0.369→1.287
MDM	Real	0.462	0.996	0.839	0.000→1.311

2,048 training motions (μ_t, Σ_t), and set per-clip targets $\mu_t + (e_i - \mu_s)T$ with the Gaussian transport map [13]

$$T = \Sigma_s^{-1/2}\left(\Sigma_s^{1/2}\Sigma_t\Sigma_s^{1/2}\right)^{1/2}\Sigma_s^{-1/2}, \qquad (1)$$

with eigenvalues floored at $10^{-4}\,\mathrm{tr}(\Sigma)/512$. Coefficients minimize squared distance to these targets for 80 Adam steps. No caption content, test reference or test-time score enters the optimization, and all outputs are saved before reference scoring. Two bounds, 0.1 and 0.35, were fixed in advance; the larger raises FID in every cohort (Tab. S6).

Query-only recoding. The attacker receives one scalar per query: the FID of its recoded outputs against the 2,048 training-motion embeddings of the primary evaluator. It sees no gradients, per-clip embeddings, captions or test reference. A (1+1) evolution strategy starts at the identity, proposes Gaussian steps of initial size $0.3b$ (multiplied by 1.5 after a success and 0.9 after a failure) and keeps a proposal only if it lowers the queried FID. It searches either 3 global coefficients or 3 coefficients per joint (63), shared across frames and clips. The best coefficients after a given number of queries are then scored once against the withheld test reference, with the primary evaluator and with the second official checkpoint, which the attacker never queried. Positions, velocities and contacts are bitwise unchanged, and canonical embeddings change by less than 10^{-6}.

Table S6. **Caption-blind FID**, 960 outputs per cohort. *Source*: evaluator whose gradients were used; *other*: the second official T2M checkpoint, which supplied no gradients. Canonical-gradient controls stay at the original value.

Cohort	Evaluator	Original	$b = 0.1$	$b = 0.35$
MDM seed 42	source	0.384	0.210	1.085
	other	0.408	0.334	0.972
MDM seed 4242	source	0.376	0.176	1.148
	other	0.371	0.303	0.856
MDM seed 7777	source	0.327	0.199	1.429
	other	0.379	0.284	0.779
MLD	source	0.706	0.289	1.611
	other	0.659	0.579	1.345

With per-joint coefficients and $b = 0.1$, the queried FID falls in every cohort and the fall carries over to the unqueried checkpoint in every cohort (Tab. S7). Most of the gain is reached within 100 queries. The larger bound lowers the queried FID further but overshoots for the other checkpoint, as in the gradient-based caption-blind attack. Three global coefficients are much weaker. R-Precision stays within 0.41–0.44 throughout.

S6. Reward fine-tuning

We fine-tune two generators against a T2M checkpoint. All settings were fixed on 512 validation clips before a single test inference on 2,048 test clips; the dated protocol, with its amendments, is released with the code. Table S8 summarizes the test results.

Table S7. **Query-only recoding**, test FID after 300 queries (relative change from the original). *Queried*: the evaluator that answered the queries; *other*: the second checkpoint, never queried. *Random*: mean of ten random coefficient vectors of the same bound on the queried evaluator.

Search		MDM 42	MDM 4242	MDM 7777	MLD
Original FID (queried / other): 0.384 / 0.408, 0.376 / 0.371, 0.327 / 0.379, 0.706 / 0.659					
Per joint, $b = 0.1$	queried	-20%	-27%	-11%	-57%
	other	-30%	-4%	-22%	-41%
	random	$+2\%$	$+3\%$	$+4\%$	$+4\%$
Per joint, $b = 0.35$	queried	-29%	-44%	-30%	-68%
	other	$+51\%$	$+12\%$	$+57\%$	$+76\%$
	random	$+89\%$	$+74\%$	$+97\%$	$+64\%$
Global, $b = 0.1$	queried	-9%	-11%	-3%	-30%
	other	$+47\%$	$+31\%$	$+28\%$	-18%

Method. Following ReFL [14] and DRaFT [15], each step samples motions for training captions without gradients down to a random step among the last 11 of 50, predicts x_0 there with gradients, and feeds it, raw or canonicalized, to a frozen T2M checkpoint. The reward loss is a symmetric InfoNCE over negative Euclidean distances between motion and caption embeddings, matching how R-Precision ranks captions. MDM [5] (guidance 2.5) also uses its diffusion loss on real training motion and a proximity penalty to the frozen pretrained model [16] (weight 100), both computed on canonicalized motion, so that only a raw reward acts on the redundant coordinates. MLD [8] (DDIM-50, guidance 7.5) uses its own latent-space terms instead, which do not single out these coordinates; only its denoiser is trained, and its VAE decoder stays frozen. Both use AdamW, learning rate 10^{-5}, batch 32, 2,000 steps and three seeds; the control drops the reward.

Selection. A setting qualified if, on validation, the reward checkpoint's raw R-Precision rose by at least 0.03 while its raw FID stayed within 1.5 times the starting value. For MDM, runs without the proximity penalty or with weight 10 failed the FID limit, so the weight was set to 100; without the penalty every reward run drifted, whichever input the reward read. At weight 100 the raw-reward version exceeded the limit only when scored raw (0.660 against 0.62; 0.244 canonicalized), which is the effect under study. The replication with the T2M-GPT checkpoint as reward reuses every setting. For MLD, latent penalties of 10 to 1,000 left the reward inactive (R-Precision gain at most 0.019), so the grid was extended to 1 and 0 before any test inference; the fixed rule, the smallest qualifying weight, selected 0. Weight 1, which also qualified, gives the same picture on validation (off-canonical 0.051 and 0.050 for raw and canonical reward).

Results. With raw reward, MDM moves off canonical form in every run, and canonicalization lowers the reward checkpoint's R-Precision by 0.008 to 0.009 with both checkpoints; canonical-reward runs stay at the pretrained level. The FID effect depends on the checkpoint. With MDM's checkpoint as reward, the same outputs score an FID of 0.75 raw and 0.28 canonicalized, against 0.33 for the pretrained model. With T2M-GPT's as reward, its own gap varies in sign across seeds ($+0.053$, -0.015, -0.132), while MDM's checkpoint, as judge, consistently rates the raw outputs better than their canonical form. MLD shows no difference between the two reward inputs: both versions drift equally and acquire the same positive FID gap, which the pretrained model and the control do not have. We did not test why; one possibility is that MLD reaches the features only through a decoder trained on canonical motion and kept frozen.

A released reward-fine-tuned model. MotionRFT [17] releases an MLD fine-tuned with the authors' own learned reward, together with its base checkpoint. Over five native cohorts of 1,024 outputs (generation seeds 2101–2105, DDIM-50, guidance 7.5; paired t_4 intervals, Bonferroni-adjusted over eight endpoints), the off-canonical distance is 0.044–0.046 for the base and 0.049–0.051 for the fine-tuned model. The raw-minus-canonical FID gap is -0.016 [-0.039, 0.007] and -0.003 [-0.010, 0.004] for the base (MDM and T2M-GPT checkpoints), against $+0.036$ [0.028, 0.045] and $+0.028$ [0.008, 0.048] after fine-tuning. As in our MLD study, we do not attribute this to the reward reading rotations; it shows that fine-tuning changed how much the score depends on the encoding, a change that canonical evaluation removes.

S7. Sign language

Data and readers. We use the released SignSparK body and hand rotation features [18] (240 per frame: 10 body and 2×15 hand joints) and the public backtranslation readers [19], one Sign Language Transformer [20] per corpus, with native tokenization, beam search and 300-frame cap. Full test sets contain 642 PHOENIX-2014T [21],

Table S8. **Reward fine-tuning**, 2,048 test captions, three seeds per version. *Off-canonical*: RMS distance of the 6D rotation numbers from their canonical form (range over seeds). The other columns are raw-minus-canonical gaps of the same outputs, by the checkpoint that gave the reward and by the other one; means over seeds with 95% intervals from 2,000 paired bootstrap resamples of clips, which do not include seed-to-seed variation. Pretrained MDM has one model, so no interval is given.

Model	Off-canonical	ΔR1, reward ckpt.	ΔFID, reward ckpt.	ΔFID, other ckpt.
MDM, reward from the MDM checkpoint				
Pretrained	0.019	−0.002	+0.005	+0.001
No reward	0.018–0.019	+0.001 [0.000, 0.002]	+0.003 [0.001, 0.005]	+0.008 [0.004, 0.013]
Canonical reward	0.019	−0.001 [−0.002, 0.001]	−0.044 [−0.076, −0.015]	+0.008 [−0.002, 0.018]
Raw reward	0.042–0.046	+0.008 [0.002, 0.014]	+0.467 [0.278, 0.658]	+0.035 [−0.035, 0.109]
MDM, reward from the T2M-GPT checkpoint				
Pretrained	0.019	0.000	+0.001	+0.005
No reward	0.018–0.019	+0.002 [0.001, 0.003]	+0.008 [0.004, 0.013]	+0.003 [0.001, 0.005]
Canonical reward	0.018–0.019	+0.001 [0.000, 0.002]	−0.004 [−0.007, −0.001]	+0.004 [0.001, 0.007]
Raw reward	0.039–0.040	+0.009 [0.003, 0.016]	−0.031 [−0.118, 0.058]	−0.051 [−0.074, −0.027]
MLD, reward from the MDM checkpoint				
Pretrained	0.044	0.000 [−0.003, 0.003]	−0.019 [−0.030, −0.010]	−0.011 [−0.023, 0.002]
No reward	0.043	−0.001 [−0.003, 0.002]	−0.021 [−0.030, −0.012]	−0.013 [−0.024, 0.000]
Canonical reward	0.059–0.061	+0.001 [−0.002, 0.003]	+0.033 [0.014, 0.053]	+0.050 [0.028, 0.073]
Raw reward	0.060–0.062	+0.003 [0.000, 0.006]	+0.031 [0.011, 0.051]	+0.045 [0.022, 0.069]

2,183 How2Sign [22] and 1,162 CSL-Daily [23] clips; real-motion BLEU is 15.00, 3.48 and 6.89. Left-hand channels, which the released pipeline preprocesses canonically, are unchanged in every attack.

S7.1. Real-input stability controls

We decode the same full How2Sign and CSL-Daily test inputs under canonicalization, Gaussian noise and physical rotation changes, keeping input order, crops, reader weights, vocabularies and beam settings fixed. Every path shares native preprocessing. The reader receives only motion and valid lengths; captions are used only after decoding to compute BLEU. Canonicalization changes 1,438 How2Sign and 464 CSL-Daily hypotheses, while all 642 PHOENIX hypotheses remain unchanged.

Controls. Each clip's control is scaled to its own canonicalization displacement $B = \|\mathcal{C}(x) - x\|_2$ over valid frames. The Gaussian control averages three fixed noise fields on the body and right-hand channels, each rescaled to B; a fourth field perturbs all 240 channels. The physical control rotates both stored vectors of each joint about a per-joint axis fixed through time, by the per-clip angle that reaches the same B (mean $2.6°$ on How2Sign and $5.3°$ on CSL-Daily); a second physical control uses $1°$. No control uses captions or gradients, and the preprocessed left hand is never changed.

Measurements. Changed decodings compare strings; normalized edit distance (NED) divides the Levenshtein distance, in words on How2Sign and characters on CSL-Daily, by the longer output. Intervals use 20,000 paired cluster bootstraps over the 149 official How2Sign videos or the CSL-Daily clips, adjusted over twenty endpoints; canonical-minus-control contrasts form a separate family of eight. They condition on the readers, inputs and noise fields.

Results. Table S9. Canonicalization changes more decodings than same-support Gaussian noise in both corpora: by $+24.0$ points [21.0, 26.7] on How2Sign and $+17.6$ [13.9, 21.4] on CSL-Daily, with NED differences of $+0.154$ [0.133, 0.174] and $+0.136$ [0.106, 0.167]. Matched physical changes move the decoding slightly more than canonicalization: -4.4 points $[-7.6, -1.3]$ and -4.4 $[-9.0, 0.2]$, NED -0.030 $[-0.053, -0.006]$ and -0.042 $[-0.080, -0.005]$. The readers are therefore sensitive to the encoding beyond what noise of the same size explains, and also to physical changes; neither result implies better or worse semantic accuracy. Results for each noise field are released with the code.

S7.2. Learned and per-clip recodings

Learned layers. The PHOENIX layer (299k parameters) reads a frozen text-to-motion diffusion model's output and the source sentence, encoded by the generator's frozen text encoder, and predicts bounded coefficients with cross-attention and three dilated temporal convolutions. It is trained for 20 epochs on 7,096 training motions with the reader's caption cross-entropy; generator and reader are frozen and final checkpoints are used. The How2Sign and CSL-Daily layers (770k parameters) read real training motion and the reader's own word embeddings. Table S10 reports ablations: without the source sentence the gain is small, and with the loss computed after canonicalization it vanishes, as Eq. (4) of the main paper predicts.

Table S9. **Real-sign stability controls**, full test sets, same frozen readers and inputs. *Changed*: decoded sentences that differ from the raw input's; NED: normalized edit distance to the raw decoding. Gaussian: mean of three same-support noise fields rescaled to each clip's canonicalization displacement; *all 240*: noise on every channel. Physical: real joint rotations, matched in input norm or fixed at 1°. Intervals: 20,000 paired cluster bootstraps, adjusted over twenty endpoints. Raw BLEU is 3.48 (How2Sign) and 6.89 (CSL-Daily).

Corpus	Condition	Changed %	95% interval	NED	95% interval	BLEU
How2Sign	Canonical	65.9	[61.7, 69.8]	0.397	[0.366, 0.429]	3.48
	Gaussian	41.9	[38.9, 45.0]	0.243	[0.223, 0.264]	3.51
	Gaussian, all 240	37.2	[33.8, 40.8]	0.220	[0.197, 0.244]	3.58
	Physical, matched	70.3	[66.6, 74.1]	0.426	[0.396, 0.458]	3.55
	Physical, 1°	45.4	[41.9, 49.1]	0.252	[0.228, 0.278]	3.60
CSL-Daily	Canonical	39.9	[35.5, 44.2]	0.304	[0.269, 0.339]	7.07
	Gaussian	22.3	[19.6, 25.2]	0.168	[0.147, 0.191]	6.74
	Gaussian, all 240	20.5	[17.0, 24.2]	0.156	[0.128, 0.186]	6.95
	Physical, matched	44.3	[40.0, 48.7]	0.346	[0.312, 0.382]	6.58
	Physical, 1°	13.9	[10.8, 17.0]	0.102	[0.078, 0.125]	7.01

Table S10. **Learned-layer ablations**, full test sets. Raw BLEU and gain over no layer with paired 95% intervals; canonical hypotheses are unchanged in every row. How2Sign/CSL controls use one seed.

Corpus	Layer input / loss	Raw	Gain [95% CI]
PHOENIX	text+motion / raw	15.21	8.81 [7.82, 9.87]
	motion only / raw	7.60	1.21 [0.66, 1.76]
	text+motion / canonical	6.26	-0.13 [-0.32, 0.04]
How2Sign	text+motion / raw	6.11	2.64 [2.27, 3.03]
	text+motion / canonical	3.48	0.00 [-0.18, 0.20]
CSL-Daily	text+motion / raw	7.43	0.54 [0.14, 0.97]
	text+motion / canonical	6.79	-0.10 [-0.40, 0.09]

Per-clip attacks. On 128 real test clips per corpus, 80 Adam steps toward a circularly shifted sentence raise BLEU against that sentence to 85.19 (PHOENIX), 47.34 (How2Sign) and 32.44 (CSL-Daily), while BLEU against the actual sentence drops to 0.00, 0.00 and 1.26; canonical hypotheses are unchanged. A gloss-level CTC recognizer of a different architecture, which we trained ourselves and audited on the PHOENIX development set, shows the same failure (wrong-target WER 118%→4%).

Detector statistics. Write $s = (\rho - 1, \sigma - 1, \tau)$. The squared distance to the canonical encoding is $E = \|s\|^2$, and the squared feature norm satisfies $N - E = 2(\rho+\sigma) - 2$, so rotating s about $(1, 1, 0)/\sqrt{2}$ preserves both. A per-clip attack restricted to this circle (200 steps) raises PHOENIX BLEU from 6.62 to 10.72 (+4.09, 95% CI [3.04, 5.24]), while a threshold detector on the RMS distance to canonical has AUC 0.500. Fixing the full 2×2 Gram matrix of (a, b) would leave no channel; matching a few statistics is therefore not a general defense, whereas canonicalization closes the whole fiber.

Reader adaptation. Real How2Sign and CSL-Daily features are not canonical. We fine-tune each public reader for five epochs on raw or on canonical training features, with three seeds each and matched batches. Test BLEU is 3.37 raw versus 3.47 canonical on How2Sign (difference $+0.10$, 95% CI [-0.01, 0.21]) and 7.24 versus 7.22 on CSL-Daily (-0.02, [-0.19, 0.15]). Canonical readers keep every hypothesis under fresh attacks and under fixed equivalent recodings of all 10,035 test inputs.

S8. Statistics and reproducibility

Motion audits use paired t_{19} intervals over 20 native draws, Bonferroni-adjusted within each stated family. Sign-language intervals use 20,000 paired bootstrap resamples that recompute corpus BLEU from sufficient statistics, clustered by the 149 official How2Sign video identifiers; seed sensitivity additionally resamples the three fitting seeds. Corpus BLEU and WER are recomputed with independent implementations. Clip selections, seeds and bounds were fixed before scoring, and no checkpoint was selected on test scores. Experiments ran with PyTorch 2.11 and CUDA 12.8 on one RTX PRO 6000; exact floating-point decoding can depend on hardware.

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